

Implicitly constrained learning: A functional approach to differentiable programming in machine learning

Oberwolfach Workshop 2613

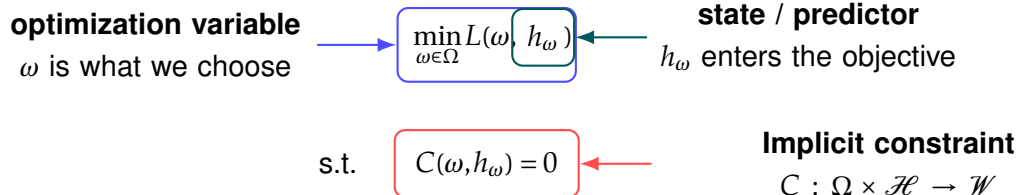
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Implicitly Constrained Learning (ICL)



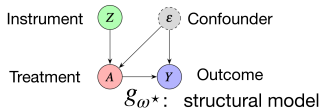
Main Goal: Gradient methods for ICL that can leverage deep learning

- ▶ **Main assumption:** h_ω is **uniquely** defined by the constraint.
- ▶ $\mathcal{H}$ a function space (e.g. Hilbert, Banach) approximated by neural networks.
- ▶ **The central object:** The implicit map $\omega \mapsto h_\omega \in \mathcal{H}$.
- ▶ Constraint and objective may be data-dependent: statistical errors.

A few applications of ICL

Causal effect estimation

Fit ω using variation induced by Z



$$\min_{\omega} \mathbb{E} [\|Y - h_{\omega}(Z)\|^2] \leftarrow L(\omega, h_{\omega})$$

Instrument-induced
Projection h_{ω}

Project $g_{\omega}(A)$ onto functions of Z

$$h_{\omega} = \arg \min_h \mathbb{E} [\|h(Z) - g_{\omega}(A)\|^2]$$

Constraint $C(\omega, h_{\omega}) = 0$

Inference under PDE constraints

Infer the initial vorticity field u_{ω}
from sparse observations at time T

$$\min_{\omega} \mathbb{E} [\|h_{obs}(x) - h_{\omega}(T, x)\|^2]$$



observed field



predicted field

Predicted field
at time T h_{ω}

Solve Navier-Stokes with known forcing f
and inferred initial vorticity u_{ω}

$$\begin{aligned} \partial_t h_{\omega} + v_{\omega}^{\top} \nabla h_{\omega} - \Delta h_{\omega} &= f \\ \text{div}(v_{\omega}) &= 0, \quad h_{\omega} = \nabla \times v_{\omega} \\ h_{\omega}(0, x) &= u_{\omega}(x) \end{aligned}$$

Planning and control

Learn a policy a_{ω} that drives
the system to a chosen goal state

$$\min_{\omega} \mathbb{E} [\|g - h_{\omega}(T)\|^2]$$



Goal state



Predicted state

Predicted state h_{ω}

Predict future states using known
dynamics model $\mathcal{M}$

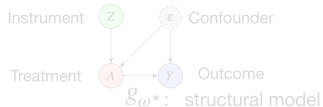
$$\frac{d}{dt} h_{\omega}(t) = \mathcal{M} \left(h_{\omega}(t), a_{\omega}(t, h_{\omega}(t)) \right)$$



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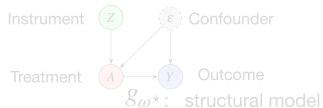
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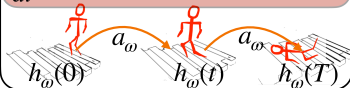


Predicted state

Predicted state h_{ω}

Predict future states using known
dynamics model $\mathcal{M}$

$$\frac{d}{dt} h_{\omega}(t) = \mathcal{M} \left(h_{\omega}(t), a_{\omega}(t, h_{\omega}(t)) \right)$$



Why deep learning enters the picture

Causal effect estimation



PDE-constrained learning



Control problems



high-dimensional covariates, spatial fields, control trajectories

Deep neural networks are remarkably effective models for high-dimensional structured data.

How can we leverage these models for solving ICL problems?

Restricted setting: Bilevel optimization

$$\min_{\omega \in \Omega} L(\omega, h_{\omega})$$

s.t.

$$h_{\omega} = \operatorname{argmin}_{h \in \mathcal{H}} G(\omega, h)$$

Optimality condition

$h \mapsto G(\omega, h)$ strongly convex

This talk:

- **Bilevel optimization:** ICL with optimality constraint:

$$C(\omega, h_{\omega}) = 0 \iff h_{\omega} = \operatorname{argmin}_{h \in \mathcal{H}} G(\omega, h)$$

- **Main application:** Causal effect estimation.

The direct approach: Parametrize then differentiate

1. Replace $\mathcal{H}$ by a parametric family $(H_\theta)_{\theta \in \Theta}$ (e.g. neural networks)
2. Solve the resulting constrained problem.

$$\begin{array}{ll} \min_{\omega \in \Omega} L(\omega, h_\omega) & \leftarrow H_{\theta(\omega)} \\ \text{s.t.} & h_\omega = \arg \min_{h \in \mathcal{H}} G(\omega, h) \quad \leftarrow \theta(\omega) \in \arg \min_{\theta} G(\omega, H_\theta) \end{array}$$

Analogous to **discretize then optimize** in PDE optimization [Hinze et al., 2008].

The direct approach: Parametrize then differentiate

1. Replace $\mathcal{H}$ by a parametric family $(H_\theta)_{\theta \in \Theta}$ (e.g. neural networks)
2. Solve the resulting bilevel optimization problem using a gradient method.

$$\min_{\omega \in \Omega} L(\omega, H_{\theta(\omega)})$$

Objective depends on choice of minimizer

s.t.

$$\theta(\omega) \in \arg\min_{\theta} G(\omega, H_\theta)$$

Minimizer not necessarily unique

- Implicit dependence on ω through the solution map $\omega \mapsto \theta(\omega)$: may be non-unique.

Ambiguously defined bilevel optimization problem

- General bilevel problems are harder than NP-hard [Bolte et al., 2024].

The direct approach: Parametrize then differentiate

$$\min_{\omega \in \Omega} L(\omega, H_{\theta(\omega)})$$

s.t.

$$\theta(\omega) \in \arg \min_{\theta} G(\omega, H_{\theta})$$

Main idea: chain rule + approximation of $\nabla_{\omega} \theta(\omega)$ (when it makes sense)

- ▶ **Algorithm unrolling:** Differentiate w.r.t. ω through the optimization procedure for θ .
 - ▶ Implicit bias of the optimization [Vicol et al., 2021].
 - ▶ Convergence guarantees when the optimum $\theta(\omega)$ is unique [Ji et al., 2020].
- ▶ **Implicit differentiation:** Implicit function theorem (IFT) to compute $\nabla \theta(\omega)$.
 - ▶ Convergence guarantees when the optimum $\theta(\omega)$ is unique [Liu et al., 2021, Arbel and Mairal, 2022a, Chen et al., 2024].
 - ▶ Extension to non-unique solutions under a restricted geometric property [Arbel and Mairal, 2022b].

Uniqueness of the minimizer $\theta(\omega)$ is key
Unrealistic when H_{θ} is an over-parameterized network!

Alternative approaches

$$\min_{\omega \in \Omega} L(\omega, h_\omega)$$

s.t.

$$h_\omega = \arg \min_{h \in \mathcal{H}} G(\omega, h)$$

- ▶ **Penalization methods:** Replace the optimality constraint by a penalty term.
 - ▶ Overfitting when L and G estimated using finite data + expressive approximating family H_θ [Krishnapriyan et al., 2021, Vicol et al., 2021].
- ▶ **Neural Operators:** Approximate the solution map $\omega \mapsto h_\omega = \arg \min_{h \in \mathcal{H}} G(\omega, h)$ with a single model network $\omega \mapsto \mathcal{T}_\theta(\omega)$ from training pairs $(\omega_i, h_{\omega_i})_{1 \leq i \leq N}$.
 - ▶ Computationally expensive [Navaneeth et al., 2024]
 - ▶ Unreliable predictions beyond training range [Zhu et al., 2023]

Can we do better?

Functional Bilevel Optimization for Machine Learning

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Neural Implicit Differentiation: Differentiate then parameterize

$$\min_{\omega \in \Omega} J(\omega) = L(\omega, h_\omega)$$

s.t.

$$h_\omega = \arg \min_{h \in \mathcal{H}} G(\omega, h)$$

Main idea: gradient w.r.t. ω by chain rule + approximation of $\nabla_\omega h_\omega$.

1. Compute an *abstract* expression of $\nabla_\omega J(\omega)$ by implicit differentiation in $\mathcal{H}$.
2. Use the neural network family $(H_\theta)_{\theta \in \Theta}$ only as a function approximation tool.

Never compute the sensitivity of the network parameter θ w.r.t. ω !

Similar to **optimize then discretize** approaches in PDE optimization [Hinze et al., 2008].

Implicit differentiation in an abstract Hilbert space $\mathcal{H}$

$$\min_{\omega \in \Omega} J(\omega) = L(\omega, h_\omega)$$

s.t.

$$h_\omega = \arg \min_{h \in \mathcal{H}} G(\omega, h)$$

Theorem (informal): Assume that $h \mapsto G(\omega', h)$ is μ -strongly convex for any $\omega \in \mathbb{R}^d$,

- ▶ G and L have finite values and are Fréchet (strongly) differentiable.
- ▶ $\partial_2 G$ is **Hadamard differentiable** (not necessarily Fréchet differentiable).

Then, J is differentiable, and : $\nabla J(\omega) = \partial_1 L(\omega, h_\omega) + \nabla h_\omega \partial_2 L(\omega, h_\omega)$.

The Jacobian ∇h_ω is the unique solution to the linear system:

$$\nabla h_\omega \overbrace{\partial_{2,2}^2 G(\omega, h_\omega)}^{\text{Operator in } \mathcal{H}} = - \overbrace{\partial_{1,2}^2 G(\omega, h_\omega)}^{\text{Operator from } \mathcal{H} \text{ to } \mathbb{R}^d}$$

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- ▶ Standard versions of IFT require $\partial_2 G$ to be **Fréchet differentiable** → **Too restrictive**
A twice Fréchet function $G(\omega, h) = \int \ell(\omega, h(x)) dx$ in L_2 is quadratic [Nemirovski and Semenov, 1973]
- ▶ **Hadamard differentiability** allows a **broad**er class of functions!
(also used in statistics for the functional delta-method [van der Vaart and Wellner, 1996].)

Implicit differentiation in an abstract Hilbert space $\mathcal{H}$

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- ✓ Expression of the gradient is **independent** of the way h_ω is approximated.
- ✗ Abstract expression, unclear how to use it in practice.

Need to consider more concrete spaces!

Implicit differentiation in L_2 spaces and adjoint sensitivity method

A notable setting in ML (e.g.: Model-based RL, Instrumental Variables regression)

- ▶ Outer and inner losses are expectations of point-wise losses under data distribution $\mathbb{P}$:

$$L(\omega, h) := \mathbb{E}_{\mathbb{P}} [\ell(\omega, h(x), y)], \quad G(\omega, h) := \mathbb{E}_{\mathbb{P}} [g(\omega, h(x), y)]$$

- ▶ $\mathcal{H} = L_2(\mathbb{P})$: The space of functions of x that are square integrable w.r.t. $\mathbb{P}$.

Proposition (informal): If $v \mapsto g(\omega, v, y)$ is strongly convex + mild assumptions:

$$\nabla J(\omega) = F(\omega, h_{\omega}, a_{\omega}) = \mathbb{E}_{\mathbb{P}} [\partial_1 \ell(\omega, h_{\omega}(x), y) + \partial_{1,2} g(\omega, h_{\omega}(x), y) a_{\omega}(x)],$$

where the adjoint function a_{ω} is the unique minimizer in $L_2(\mathbb{P})$ of $a \mapsto L_{adj}(\omega, h_{\omega}, a)$:

$$L_{adj}(\omega, h, a) := \mathbb{E}_{\mathbb{P}} \left[\frac{1}{2} a(x)^{\top} \partial_{2,2}^2 g(\omega, h(x), y) a(x) + a(x)^{\top} \partial_2 \ell(\omega, h(x), y) \right]$$

Neural Implicit Differentiation

$$\begin{aligned} \min_{\omega \in \Omega} J(\omega) &= L(\omega, h_\omega) & \text{s.t.} & \quad h_\omega = \arg\min_{h \in \mathcal{H}} G(\omega, h) \\ \nabla_\omega J(\omega) &= F(\omega, h_\omega, a_\omega) & \text{with} & \quad a_\omega = \arg\min_{a \in \mathcal{A}} L_{adj}(\omega, h_\omega, a) \end{aligned}$$

Algorithm:

Input: ω_0 , Networks $(H_\theta)_{\theta \in \Theta}$ and $(A_\xi)_{\xi \in \Xi}$

for $t = 0, \dots, T - 1$ **do**

Inner-level optimization

$$\hat{\theta} \approx \arg\min_{\theta} \widehat{G}(\omega_t, H_\theta)$$

Use $H_{\hat{\theta}}$ to approximate h_{ω_t} .

Adjoint optimization

$$\hat{\xi} \approx \arg\min_{\xi} \widehat{L_{adj}}(\omega_t, H_{\hat{\theta}}, A_\xi)$$

Use $A_{\hat{\xi}}$ to approximate a_{ω_t} .

Gradient estimation

$$\omega_{t+1} \leftarrow \omega_t - \gamma \widehat{F}(\omega_t, H_{\hat{\theta}}, A_{\hat{\xi}});$$

end for

- ▶ **Memory and time savings:** No need for higher-order derivatives of H_θ and A_ξ .
- ▶ **Stability:** Strongly-convex adjoint objective in function space: well-defined solution.

Convergence/generalization guarantees for NID

Proposition (informal): Assume that J is L -smooth and admits a finite lower bound J^* . Consider the sequence ω_{t+1} produced by the NID algorithm with suitable step-size η . Under mild smoothness assumptions on the objectives and data:

$$\min_{0 \leq t \leq T} \mathbb{E} \left[\|\nabla J(\omega_t)\|^2 \right] \leq \frac{4(J(\omega_0) - J^*)}{\eta(t+1)} + 2\eta L \sigma_{\text{eff}}^2 + (c_1 \epsilon_{\text{in}} + c_2 \epsilon_{\text{adj}}),$$

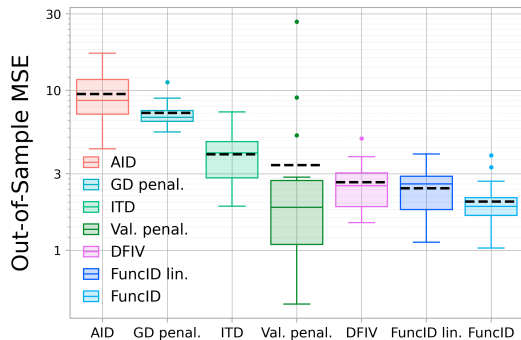
where $c_1, c_2, \sigma_{\text{eff}}^2$ are positive constants, and $\epsilon_{\text{in}}, \epsilon_{\text{adj}}$ are sub-optimality errors that result from the inner and adjoint optimization procedures.

- ▶ Sub-optimality errors are hard to control for neural networks: depends on optimization, approximation power, complexity of the class.
- ▶ The rate does not quantify the effect of sample size: (all hidden in the sub-optimality error).

Hard to get quantitative convergence results in L_2 -spaces, what about other spaces?

Applications to Causal effect estimation

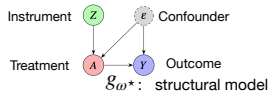
Results on synthetic task from [Xu et al., 2020]



- ▶ Large improvement over classical bilevel optimization algorithms (AID/ITD)
- ▶ Competitive with problem specific methods (DFIV)!

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Constraint $C(\omega, h_{\omega}) = 0$

Learning Theory for Kernel Bilevel Optimization

Fares El Khoury¹ Edouard Pauwels² Samuel Vaiter³ Michael Arbel¹



Reproducing Kernel Hilbert Spaces meet Bilevel optimization

Kernel Bilevel optimization

$$\begin{aligned} \min_{\omega \in \Omega} J(\omega) &:= L(\omega, h_{\omega}^{\star}) \\ \text{s.t. } h_{\omega}^{\star} &\in \operatorname{argmin}_{h \in \mathcal{H}} G(\omega, h) \end{aligned}$$

$$\begin{aligned} L(\omega, h) &= \mathbb{E}_{\mathbb{P}} [\ell(\omega, h(x), y)], \\ G(\omega, h) &= \mathbb{E}_{\mathbb{P}} [g(\omega, h(x), y)] + \frac{\lambda}{2} \|h\|_{\mathcal{H}}^2, \end{aligned}$$

- Same as before, but now $\mathcal{H}$ is an RKHS with r.k. K .
- Already appeared in the past: [Keerthi et al., 2006, Kunapuli et al., 2008].

Why an RKHS?

- ▶ **Expressiveness:** Some RKHS are dense in L_2 spaces [Steinwart and Christmann, 2008].
- ▶ **Algorithms:** Easy to derive thanks to the *Representer theorem* (next slide).
- ▶ **Learning theory:** Well-established theoretical framework available for regression (and similar problems) [Smale et al., 2005, Caponnetto and De Vito, 2007].

A learning theory for KBO: Main results

Maximal inequalities

Theorem (informal): Fix a compact subset Ω of $\mathbb{R}^d$. Then, under mild assumptions:

$$\mathbb{E} \left[\sup_{\omega \in \Omega} \left\| \nabla \mathcal{L}(\omega) - \widehat{\nabla \mathcal{L}}(\omega) \right\| \right] \lesssim \frac{1}{\sqrt{n}},$$

Generalization error of BO algorithms

Corollary (informal): Consider the iterates $\omega_{t+1} = \omega_t - \eta \nabla \widehat{\mathcal{L}}(\omega_t)$. Then, assuming ℓ is coercive in its second argument, it follows that:

$$\mathbb{E} \left[\min_{i=0, \dots, t} \left\| \nabla \mathcal{L}(\omega_i) \right\| \right] \lesssim \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{t+1}}.$$

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Proof sketch and challenges

1 Upper-bound $\|\nabla \mathcal{L}(\omega) - \widehat{\nabla \mathcal{L}}(\omega)\|$ by terms:

$$\left\| \frac{1}{n} \sum_{i=1}^n \tau_{\omega}(x_i, y_i) - \mathbb{E}_{\mathbb{P}}[\tau_{\omega}(x_i, y_i)] \right\|,$$

where τ_{ω} can be **real-valued** functions or **RKHS-valued** ones (e.g. values in $\mathcal{H}$).

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2 **Real-valued** τ_{ω} : Maximal inequalities for empirical processes [Van der Vaart, 2000].

3 **RKHS-valued** τ_{ω} : Maximal inequalities for U -processes [Sherman, 1994].
(Applied to the squared error!)

⚠ Degraded rates when using "simpler" approaches for RKHS-valued τ_{ω} .

Summary

A new framework for bilevel optimization in ML

- ▶ Compatible with flexible function approximation tools (neural networks, RKHS).
- ▶ Practical algorithms like NID
- ▶ Opens the way to a learning theory for implicitly constrained problems.

Limitations and future work

- ▶ Extension to other spaces of functions: e.g. Sobolev spaces: learning PDEs, measure spaces [Marion et al., 2024].
- ▶ Learning theory beyond RKHS: deep networks

Thank you!



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